

Máme tedy matici přechodu

$$P = \begin{pmatrix} 1 & 3 \\ 2 & 7 \end{pmatrix} \quad \text{a} \quad \text{v\u00edne, \u017ee pro } \vec{v} \in V \quad \text{plat\u00ed} \quad \vec{v} = P[\vec{v}]_B$$

tj. $[\vec{v}] = P^{-1} \cdot \vec{v}$

$$\left(\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{array} \right) \sim \left(\begin{array}{cc|cc} 1 & 0 & 7 & -3 \\ 0 & 1 & -2 & 1 \end{array} \right)$$

$$P^{-1} = \begin{pmatrix} 7 & -3 \\ -2 & 1 \end{pmatrix}$$

Podle tvrzení 8.50 plat\u00ed, \u017ee ve v.p. V se sh. s. $\langle, \rangle$
a ON b\u00e1z\u00ed B

$$\langle \vec{u}, \vec{v} \rangle = [\vec{u}]_B^* [\vec{v}]_B \quad (\text{pro } \vec{u}, \vec{v} \in V)$$

Nad $\mathbb{R}$ je $\forall \vec{w} \in V, \vec{w}^* = \vec{w}^T$.

Tedy, pro $\vec{u} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \vec{v} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$ plat\u00ed

$$\begin{aligned} \langle \vec{u}, \vec{v} \rangle &= [\vec{u}]_B^T [\vec{v}]_B = (P^{-1} \vec{u})^T (P^{-1} \vec{v}) = (\vec{u}^T P^{-1T}) (P^{-1} \vec{v}) \\ &= \vec{u}^T (P^{-1T} P^{-1}) \vec{v} \end{aligned}$$

$$\text{Necht\u00e9} \quad A = P^{-1T} P^{-1} = \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix} \begin{pmatrix} 7 & -3 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 53 & -23 \\ -23 & 9 \end{pmatrix}.$$

Tud\u00ed\u017e

$$\begin{aligned} \left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right\rangle &= (x_1 \ x_2) \begin{pmatrix} 53 & -23 \\ -23 & 9 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \\ &= 53x_1y_1 - 23x_1y_2 - 23x_2y_1 + 9x_2y_2 \end{aligned}$$